MIL-H-46346(Ord)
15 June 1962
SUPERSEDING
FA(FC)-PD-36
20 October 1959

MILITARY SPECIFICATION

HOLDERS, TELESCOPE MOUNT: XM5 AND XM6

1. SCOPE

1.1 This specification covers two similar types of cross leveling devices known as telescope mount holders (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the Issue in effect on date of Invitation for bids form a part or this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: (General Specification (Governing the Manufacture and Inspection of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging of
MIL-I-45208	Inspection Requirements, General Specifications for
MIL-I-45607	Inspection Equipment, Supply and Maintenance for ordnance

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U. S. Military Property

FSC 1240

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:39Z
Check the source to verify that this is the current version before use.

DRAWING

ordnance Corps

F8579638	Holder, Telescope Mount: XM5
F8579639	Holder, Telescope Mount: XM6
IEL 8579638	Index or Inspect Ion Equipment List
IEL 8579639	Index of Inspection Equipment List
SQAPL 8579638	Supplementary Quality Assurance Provisions List
SQAPL 8579639	Supplementary Quality Assurance Provisions List

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model. - At the beginning of production, one holder, representative of production shall be made available to the Government to determine if the holder meets all the requirements of this specification.

3.2 Materials. - The materials shall be In accordance with the drawings, material specifications, and general specifications form ing a part of this specificaiton.

3.3 Construction. - The holders shall be manufactured in accordance with Drawing F8579638, "Holder, Telescope Mount: XM5 and Drawing F8579639, Holder, Telescope Mount: XM6" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.4 General specification.- The constractor shall be responsible for adherence to, and compilance with, the following requirements or Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish painting and protective
(e) Workmanship

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:39Z
Check the source to verify that this is the current version before use.

3.5 Environmental.

3.5.1 Operating temperatures, - The holders Shall meet the requirements special in 3.6.3 while exposed to and thermally stabilized at ambient temperatures of plus 150 degrees Fahrenheit and minus 65 degrees Fahrenheit

3.6 Performance. - All performance requirements shall be met with the holder positioned as follows:

(a) The axis of points A and C (Figure 1) shall be positioned in accordance with the reference dimensions in Figure 1.

(b) With the dimensions established in accordance with (a) above, the holder shall be positioned in a horizontal plane as shown in Figure 1, within 20 seconds.

(C) The locating surface in the dovetail (surface "X" in Figure 1) shall be adjusted parallel to a vertical plane within 20 seconds, by means of rotating the grip.

(d) The axis of "C" shall be horizontal within 20 seconds.

(e) The Spring, 8246178, shall be Positioned in accord ance with the note in Figure 1.

3.6.1 Dovetail (elevation). - The centerline passing between the taper surfaces of the dovetail shall be verticla within 3 minutes.

3.6.2 Displacement. - Rotation of the grip shall cause displacement (cont) of the dovetail (in the rocker assembly) at least 11 degrees, 15 minutes on each side of the vertical position.

3.6.3 Torque.- The torque required to rotate the grip through full travel shall not be less than 1 pound inch nor more than 12 pound inches at the standard ambient temperature (60 to 90 degrees Fahrenheit) and when measured at the temperatures specified in 3.5.1, the torque shall not have changed from the value obtained at the standard ambient temperature by more than 25 percent.

3.7 Identification marking.- Unless otherwise specified, identification marking shall be in accordance with Standard MIL-STD-130 l

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:39Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 The Supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the (government as specified In the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements. - Inspection shall be In accordance with (General Specification MIL-I-45208.

4.3 Production model. - Unless otherwise specified in the contract, one holder shall be seleted by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the (Government. This inspecion shall be conducted at the contractors facility, or Government approved laboratory, using test equipment authorized for use on the contracts. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 Production model failure.- Should a Production model fall to pass the prescribed test. The contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections In the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.4 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105

4.4.1 Defective units and lots shall be processed as specified In Standard MIL-STD-105.

4.5 Inspection of components.- Inspection requirements for component parts and subassemblies are contained herein and on Supplementary Quality Assurance Provisions (SQAP listed in Supplementary Quality Assurance Provisions List (LSQAP)-8579638 and (LSQAP)-8579639.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:39Z
Check the source to verify that this is the current version before use.

4.5.1 Contents of SQAP. - The contents of' SQAP shall represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lot of those components covered by a SQAP. The (government reserves the right, however, to Inspect for the requirements of all drawings and specifications and to reject individual non-conforming Items.

4.5.2 Contractors inspection for all components not covered by SQAP. - Inspection for these components shall be in accordance with the Inspection procedures of the written Inspection system required by specification MIL-I-45208. The system shall include procedures for checking all positional and functional characteristics shown on the drawings which Indicate precision requirements and have a bearing on the functional accuracy and performance of the holders.

4.6 Materials. - Control of material shall be in accordance with test data requirements of specification MIL-I-45208.

4.7 Inspection equipment.- Inspection equipment shall conform to drawings listed on IEL-8579638 and IEL-8579639 as proveded for under the alternative inspection procedures and inspection equipment requirement of Specification MIL-I-45208. Provisions of Specification MIL-I-45607 shall apply to test equipment provided by the Government or provided by the contractor and utilized in the Government approved inspection system. The Government reserves the right to use all approved test equipment, both Government furnished and contractor acquired, for verification inspection

4.7.1 Accuracy requirements.- Inspection equipment which incoporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of' ten percent of the specified tolerances except as otherwise specified. In cases where the tem percent accuracy imposes Impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.8 Tests.

4.8.1 Classification of defects.- The classification of defects in Tables I and shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to Inspect for any or all applicable requirements and to reject individual non-conforming items.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:39Z
Check the source to verify that this is the current version before use.

4.8.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characterstics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table IIIA with Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 2.5%	**Req't Par.**	**QA. Par.**	**Test Method**
101. **Performance**	3.6	4.9.2	**Special Testing Equipment (STE) Gage**
102. **Dovetail (elevation)**	3.6.1	4.9.3	**STE-Gage**
103. **Displacement**	3.6.2	4.9.4	**STE-Gage**
104. **Torque**	3.6.3	4.9.5	**STE-Gage**
105. **Identification marking**	3.7	4.9.6	**Visual**

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level L7 of Table IIIB with Sampling Plan Table IV of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: NONE DEFINED

MINOR: AQL 4.0%	**Req't Par.**	**QA. Par.**	**Test Method**
201. **Operating temperatures**	3.5.1	4.9.1	**Cold Chest & Oven**

4.9 Test methods and procedures.

4.9.1 Operating temperature. - Operating temperature test consists of' exposing the holders to temperatures indicated in 3.5.1 by means of a cold chest and an oven capable of maintaining these temperatures until the holders are thermally stabilized. While at these temperatures, the tests for torque requirements of 3.6.3 are made by means of a commercial torque wrench and an adapter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:39Z
Check the source to verify that this is the current version before use.

4.9.2 Performance requirements. - Performance requirement tests testing equipmen are made by means of special, Holding Fixture 764656 with Perpendicularity Gage 7646979 which will position Holder XM5 or Holder xM6 as Indicated in 3.6. The special testing device, when located on a cross leveled surface plate, can be checked for (b) and (d) requirements of 3.6 by means of a height gage located on the surface plate. The dovetail surface requirement (c) is checked by means of the perpendicularity gage located in the dovetail slot. A level is placed on the top surface of the gage perpendicular to the flat surface of' the dovetail and the grip resolved until the bubble Is within tolerance of 3.6(c). The Spring 8246178 is positioned by means of a torque wrench attached to the shaft and clamped in position by means of the two support caps of the device for requirements 3.6(e).

4.9.3 Dovetail elevation. - Dovetail elevation test consists of mounting the holders on the special testing equipment as described In 4.9.2 and locating a level on the top surface of the gage parallel to the flat surface of' the dovetail. The level bubble shall be within the tolerance of 3.6.1.

4.9.4 Displacement.- Displacement of the dovetail test is accoplished by using the same set-up as used in 4.8.2 except an optical protract is used in place Of the level. The protracting is placed on the top surface of the gage, perpendicular to the flat surface or the dovetail with dovetail in the vertical position. The grip is then rotated to the limit of movement in both directions, and at each limit the angle is measured with the protractor on the top surface of the gage. The angle on each side of the vertical position shall meet the requirements of 3.6.2.

4.9.5 Torque.- Torque test consists of measuring the torque required to rotate the rocker assembly through full travel at room temperature by means of a commercial torque wrench with a special adapter for requirements of 3.6.3. These torque values are then compared with operating temperatures test of 3.5.1 and the difference shall not exceed the tolerance of 3.6.3.

4.9.6 Identification marking. - Identiflcation marking is a visual test made in accordance with Standard MIL-STD-130.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T05:39Z
Check the source to verify that this is the currenversion before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. - The holders shall be cleaned, preserved, packaged, packed, and marked in accordance with Specification MIL-F-14619/P9279 for Telescope Mount Holder XM5 and Specification MIL-F-14619/P8579639 for Telescope Mount Holder XM6. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The holders are intended to support Elbow Telescope XM107 together with Mount Telescope XM117 and to provide cant correction to maintain the line of sight in a fixed relationship to the gun bore axis employed in the Battle Group Weapon Systems XM28 and XM29.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservations packaging and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery (see 3.1).

6.3 Design information.- Design information regarding specific equipment not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.4 Definitions.

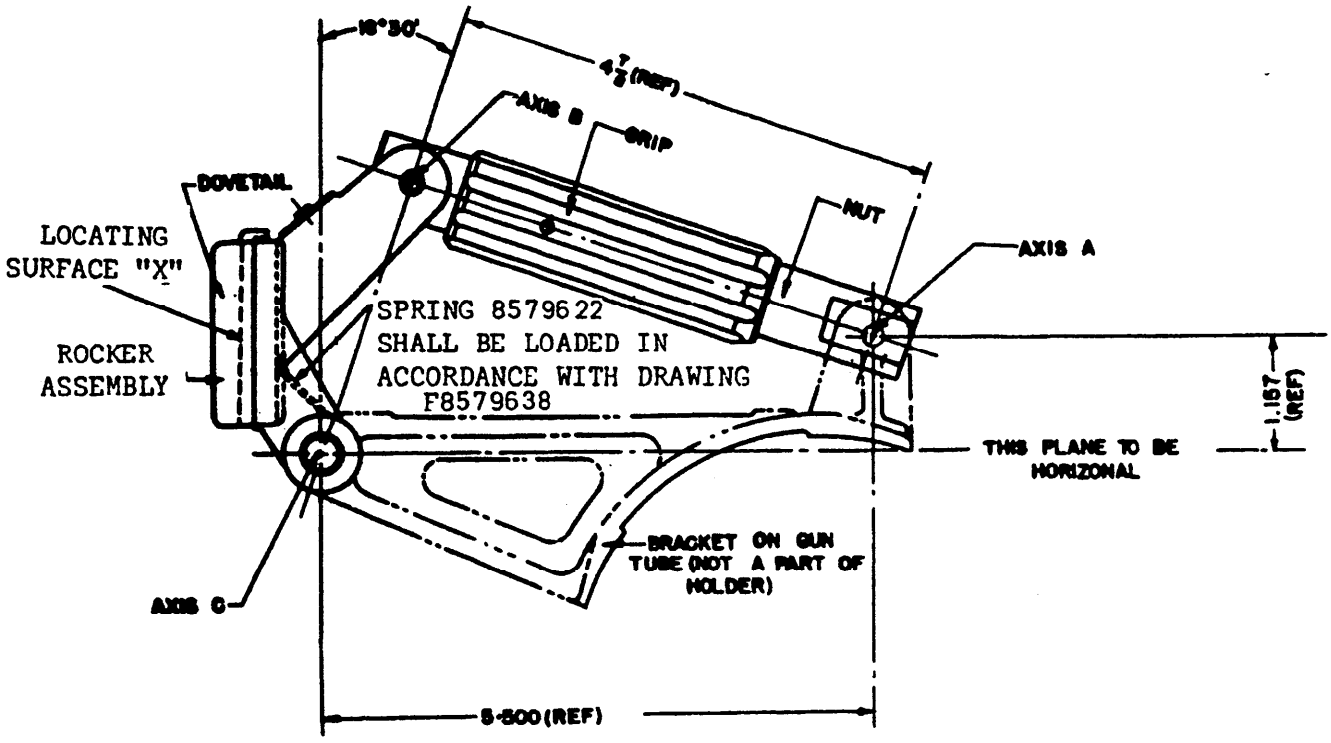
6.4.1 Special testing equipment (STE).- Special testing equipment is defined as special inspection fixtures, instruments, and modified standard measuring equipment,

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related (Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may In any way be related thereto.

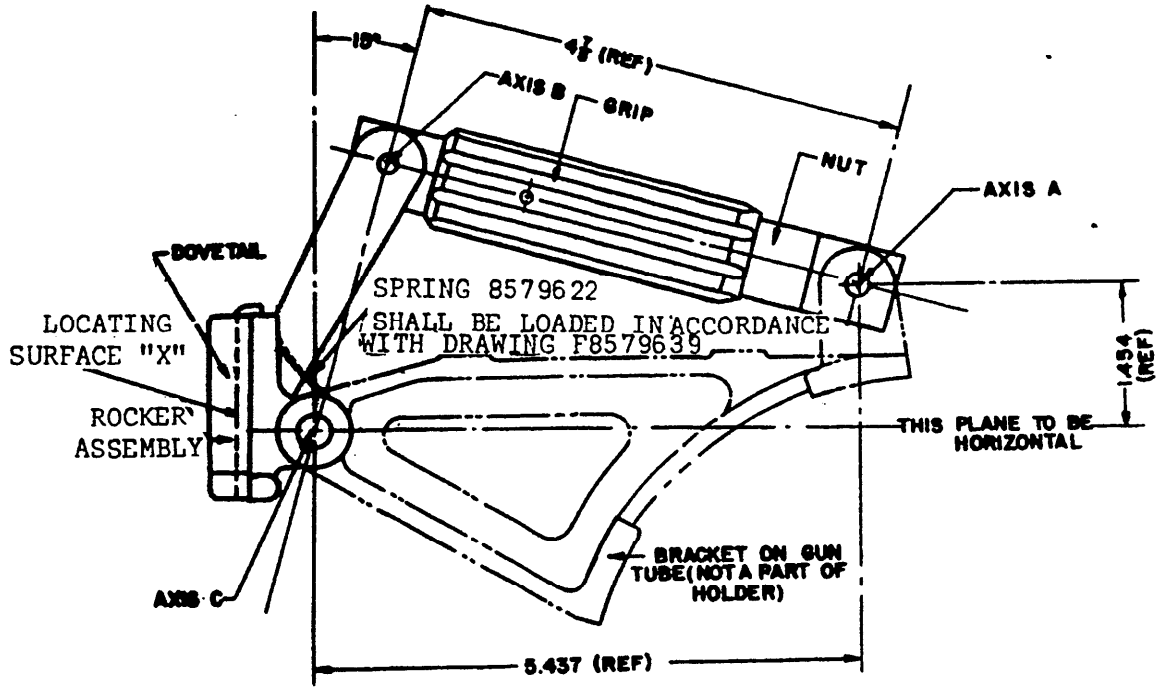
Custodian:
Army-Oral

Preparing activity:
Army-Ord

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:39Z
Check the source to verify that this is the current version before use.



(a) TELESCOPE MOUNT HOLDER, XM5



(b) TELESCOPE MOUNT HOLDER, XM6

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:39Z
Check the source to verify that this is the current version before use.